Weakest-Link Monotonicity Theorem for Majority-Vote Panel ASR

1 Setup and Definitions

Definition 1 (Panel and Majority Vote)**.** *A* panel $\mathcal{P} = \{J_1, \ldots, J_K\}$ *consists of K judges (K odd) that independently evaluate a candidate text. The panel uses* majority-vote aggregation*: it is* compromised *if and only if at least $t = \lceil K/2 \rceil$ judges are individually compromised.*

Definition 2 (Attack Model)**.** *An* attack $a \in \mathcal{A}$ *is an adversarial perturbation applied to the input that all judges evaluate. Under attack a, the judges' failure outcomes are described by a joint distribution over binary random variables $(X_1^a, \ldots, X_K^a) \in \{0,1\}^K$, where $X_j^a = 1$ indicates that judge J_j is compromised. The* individual vulnerability *of J_j under attack a is*

$$\eta_j(a) \;=\; \Pr(X_j^a = 1).$$

The panel ASR *under attack a is*

$$\mathrm{ASR}(\mathcal{P}, a) \;=\; \Pr\Big(\textstyle\sum_{j=1}^{K} X_j^a \;\geq\; t\Big).$$

Definition 3 (Adaptive Adversary)**.** *An* adaptive adversary *has full knowledge of the panel composition $\mathcal{P}$, including the per-judge vulnerability profiles $\{\eta_j(\cdot)\}_{j=1}^K$ and the inter-judge correlation structure. The adversary selects $a^* \in \mathcal{A}$ to maximize the panel ASR:*

$$\mathrm{ASR}^*(\mathcal{P}) \;=\; \sup_{a \in \mathcal{A}} \mathrm{ASR}(\mathcal{P}, a).$$

Definition 4 (Pointwise Vulnerability Dominance)**.** *Let $\mathcal{P} = \{J_1, \ldots, J_{K-1}, J_K\}$ and $\mathcal{P}' = \{J_1, \ldots, J_{K-1}, J'_K\}$ be two panels sharing the first $K-1$ judges. Write $\mathbf{X}_{-K}^a = (X_1^a, \ldots, X_{K-1}^a)$ for the shared judges' outcomes. We say J'_K is* pointwise more vulnerable *than J_K if, for every attack $a \in \mathcal{A}$ and every realization $\mathbf{x}_{-K} \in \{0,1\}^{K-1}$:*

$$\Pr\big(X_{K'}^a = 1 \;\big|\; \mathbf{X}_{-K}^a = \mathbf{x}_{-K}\big) \;\geq\; \Pr\big(X_K^a = 1 \;\big|\; \mathbf{X}_{-K}^a = \mathbf{x}_{-K}\big).$$

We further require that for each attack a, the marginal joint distribution of $\mathbf{X}_{-K}^a$ is the same in both panels $\mathcal{P}$ and $\mathcal{P}'$.

Remark. *Under conditional independence (i.e., $X_K^a \perp \mathbf{X}_{-K}^a$ for all a), pointwise dominance reduces to* marginal dominance*: $\eta_{K'}(a) \geq \eta_K(a)$ for all $a \in \mathcal{A}$. The conditional form is needed when judge failures are correlated—for instance, when a shared architectural vulnerability causes multiple judges to fail on the same inputs.*

2 Warm-Up: $K = 3$, Independent Judges

Lemma 5 ($K = 3$ Monotonicity)**.** *For a panel of $K = 3$ conditionally independent judges with vulnerabilities $\eta_1, \eta_2, \eta_3 \in [0, 1]$ under a fixed attack, the panel ASR under majority vote ($t = 2$) is*

$$\text{ASR} \;=\; \eta_1\eta_2 + \eta_1\eta_3 + \eta_2\eta_3 - 2\,\eta_1\eta_2\eta_3,$$

and is strictly increasing in each η_j whenever the other two judges are not simultaneously trivial. Letting $\eta_3 = \eta_{\max}$ without loss of generality:

$$\frac{\partial\,\text{ASR}}{\partial\,\eta_{\max}} \;=\; \eta_1(1-\eta_2) + \eta_2(1-\eta_1) \;\geq\; 0,$$

with equality if and only if $(\eta_1, \eta_2) \in \{(0, 0),\, (1, 1)\}$.

Proof. The panel is compromised when at least 2 of 3 judges fail. Under independence, enumerating the four favorable configurations:

$$\begin{aligned}\text{ASR} &= \Pr(X_1{=}1,\, X_2{=}1,\, X_3{=}0) + \Pr(X_1{=}1,\, X_2{=}0,\, X_3{=}1) \\ &\quad + \Pr(X_1{=}0,\, X_2{=}1,\, X_3{=}1) + \Pr(X_1{=}1,\, X_2{=}1,\, X_3{=}1) \\ &= \eta_1\eta_2(1{-}\eta_3) + \eta_1(1{-}\eta_2)\eta_3 + (1{-}\eta_1)\eta_2\eta_3 + \eta_1\eta_2\eta_3. \qquad (1)\end{aligned}$$

Collecting terms:

$$\begin{aligned}\text{ASR} &= \eta_1\eta_2 \underbrace{- \eta_1\eta_2\eta_3 + \eta_1\eta_3 - \eta_1\eta_2\eta_3 + \eta_2\eta_3 - \eta_1\eta_2\eta_3 + \eta_1\eta_2\eta_3}_{\text{from the last three terms}} \\ &= \eta_1\eta_2 + \eta_1\eta_3 + \eta_2\eta_3 - 2\,\eta_1\eta_2\eta_3.\end{aligned}$$

Differentiating with respect to η_3:

$$\frac{\partial\,\text{ASR}}{\partial\,\eta_3} \;=\; \eta_1 + \eta_2 - 2\,\eta_1\eta_2 \;=\; \eta_1(1-\eta_2) + \eta_2(1-\eta_1).$$

Since $\eta_1, \eta_2 \in [0, 1]$, both summands are non-negative. The derivative vanishes iff both are zero: $\eta_1(1{-}\eta_2) = 0$ and $\eta_2(1{-}\eta_1) = 0$. If $\eta_1 = 0$, the second equation forces $\eta_2 = 0$. If $\eta_2 = 1$, the first is automatic, and the second forces $\eta_1 = 1$. Hence equality holds iff $(\eta_1, \eta_2) \in \{(0, 0),\, (1, 1)\}$. □

3 General K, Independent Judges

Lemma 6 (General Monotonicity, Independent Case)**.** *For a panel of K conditionally independent judges under a fixed attack, the panel ASR under majority vote with threshold $t = \lceil K/2 \rceil$ satisfies*

$$\text{ASR} \;=\; \Pr(Y \geq t) \;+\; \Pr(Y = t{-}1) \,\cdot\, \eta_K,$$

where $Y = \sum_{j=1}^{K-1} X_j$ counts failures among the other $K{-}1$ judges. Consequently,

$$\frac{\partial\,\text{ASR}}{\partial\,\eta_K} \;=\; \Pr(Y = t{-}1) \;\geq\; 0.$$

Proof. Conditioning on the binary outcome of judge K:

$$\begin{aligned}\mathrm{ASR} &= \Pr(Y + X_K \geq t) \\ &= \Pr(Y \geq t{-}1) \cdot \eta_K \;+\; \Pr(Y \geq t) \cdot (1 - \eta_K) \\ &= \Pr(Y \geq t) \;+\; \big[\Pr(Y \geq t{-}1) - \Pr(Y \geq t)\big] \cdot \eta_K \\ &= \Pr(Y \geq t) \;+\; \Pr(Y = t{-}1) \cdot \eta_K.\end{aligned}$$

The second line uses independence of X_K and Y, conditioning on whether $X_K = 1$ (needing $Y \geq t{-}1$) or $X_K = 0$ (needing $Y \geq t$). The result is affine in η_K with slope $\Pr(Y = t{-}1) \geq 0$. □

Remark (Pivotal Voter Interpretation)**.** *The slope* $\Pr(Y = t{-}1)$ *is exactly the probability that judge* K *is* pivotal*: the remaining* $K{-}1$ *judges produce exactly* $t{-}1$ *failures, and judge* K*'s outcome determines the panel decision. Monotonicity holds because a judge can only tip the panel toward compromise (by failing), never away from it.*

4 Main Result: Correlated Judges, Adaptive Adversary

Theorem 7 (Weakest-Link Monotonicity)**.** *Let* $\mathcal{P} = \{J_1, \ldots, J_{K-1}, J_K\}$ *and* $\mathcal{P}' = \{J_1, \ldots, J_{K-1}, J'_K\}$ *be two panels using majority vote with threshold* $t = \lceil K/2 \rceil$*, sharing judges* $J_1, \ldots, J_{K-1}$ *and satisfying Definition 4 (pointwise vulnerability dominance of* J'_K *over* J_K*). Then:*

(a) For every fixed attack $a \in \mathcal{A}$*:*

$$\mathrm{ASR}(\mathcal{P}', a) \;\geq\; \mathrm{ASR}(\mathcal{P}, a).$$

(b) For any adversary strategy, including an adaptive adversary:

$$\mathrm{ASR}^*(\mathcal{P}') \;\geq\; \mathrm{ASR}^*(\mathcal{P}).$$

In particular, replacing the weakest panel member with a more vulnerable judge (increasing $\eta_{\max}$*) cannot decrease the panel ASR.*

Proof. **Part (a): Fixed attack.** Fix an arbitrary attack $a \in \mathcal{A}$. Let $\mathbf{x}_{-K} = (x_1, \ldots, x_{K-1}) \in \{0,1\}^{K-1}$ denote a realization of the shared judges' outcomes, and let $s = \sum_{j=1}^{K-1} x_j$. By the law of total probability:

$$\mathrm{ASR}(\mathcal{P}, a) \;=\; \sum_{\mathbf{x}_{-K} \in \{0,1\}^{K-1}} \Pr(\mathbf{X}^a_{-K} = \mathbf{x}_{-K}) \;\cdot\; \Pr\big(s + X^a_K \geq t \;\big|\; \mathbf{X}^a_{-K} = \mathbf{x}_{-K}\big). \tag{2}$$

Since $X^a_K \in \{0,1\}$, the inner probability depends on s in exactly three cases:

- **Case** $s \geq t$**:** The panel is compromised regardless of X^a_K. Both panels contribute 1.
- **Case** $s = t{-}1$**:** The panel is compromised iff $X^a_K = 1$ (resp. $X^a_{K'} = 1$). By pointwise dominance:
$$\Pr(X^a_{K'} = 1 \mid \mathbf{x}_{-K}) \;\geq\; \Pr(X^a_K = 1 \mid \mathbf{x}_{-K}).$$
- **Case** $s \leq t{-}2$**:** Even with $X^a_K = 1$, the total $s + 1 \leq t{-}1 < t$. Neither panel is compromised; both contribute 0.

Since the marginal distribution of $\mathbf{X}^a_{-K}$ is identical in both panels (Definition 4), summing over all realizations yields

$$\mathrm{ASR}(\mathcal{P}', a) \;\geq\; \mathrm{ASR}(\mathcal{P}, a). \tag{3}$$

Part (b): Adaptive adversary. Taking the supremum over all attacks and applying (3):

$$\mathrm{ASR}^*(\mathcal{P}') \;=\; \sup_{a \in \mathcal{A}} \mathrm{ASR}(\mathcal{P}', a) \;\geq\; \sup_{a \in \mathcal{A}} \mathrm{ASR}(\mathcal{P}, a) \;=\; \mathrm{ASR}^*(\mathcal{P}),$$

since each term in the supremum on the right is bounded by the corresponding term on the left. □

Remark (Generality of the Result)**.** *Three aspects of Theorem 7 merit emphasis:*

(i) ***Arbitrary correlations.*** *The proof makes no independence assumption. Judges may exhibit arbitrary failure correlations—including those induced by shared architectural vulnerabilities, common training data, or attack transfer effects—as long as pointwise dominance holds.*

(ii) ***Threshold generality.*** *The proof uses only the threshold structure of majority vote. It extends verbatim to any t-out-of-K aggregation rule, including unanimity ($t = K$) and any-judge-fails ($t = 1$).*

(iii) ***Adversary generality.*** *Part (b) holds for any adversary class that selects from the same attack space $\mathcal{A}$, including randomized adversaries (who select a distribution over $\mathcal{A}$, since a mixture preserves the pointwise inequality).*

Remark (Sufficient Conditions for Pointwise Dominance)**.** *Pointwise dominance (Definition 4) is satisfied whenever:*

(i) ***Model capability ordering.*** *J'_K has strictly lower capability than J_K on all tasks, so any attack that fools J_K also fools J'_K, and J'_K is vulnerable to additional attacks.*

(ii) ***Vulnerability superset.*** *The set of inputs on which J'_K errs is a superset of those on which J_K errs, for every attack.*

(iii) ***Conditional independence.*** *If $X^a_K \perp \mathbf{X}^a_{-K}$ for all a, then pointwise dominance reduces to $\eta_{K'}(a) \geq \eta_K(a)$ for all a (marginal dominance), which is empirically verifiable by measuring individual ASR.*

Remark (Coupling Interpretation)**.** *An equivalent proof uses a coupling argument: construct $(X^a_K, X^a_{K'})$ on a common probability space such that $X^a_{K'} \geq X^a_K$ almost surely (feasible by the standard coupling for stochastic dominance whenever pointwise dominance holds). Then $\sum_{j=1}^{K-1} X^a_j + X^a_{K'} \geq \sum_{j=1}^{K-1} X^a_j + X^a_K$ pointwise, so $\{$panel $\mathcal{P}$ fails$\} \subseteq \{$panel $\mathcal{P}'$ fails$\}$ and* $\mathrm{ASR}(\mathcal{P}', a) \geq \mathrm{ASR}(\mathcal{P}, a)$.

5 Tightness and Bounds

Proposition 8 (Equality Conditions)**.** *Under the conditions of Theorem 7, for a fixed attack a:*

$$\mathrm{ASR}(\mathcal{P}', a) = \mathrm{ASR}(\mathcal{P}, a)$$

if and only if, for every realization $\mathbf{x}_{-K}$ with $\sum x_j = t{-}1$ and $\Pr(\mathbf{X}^a_{-K} = \mathbf{x}_{-K}) > 0$:

$$\Pr(X^a_{K'} = 1 \mid \mathbf{x}_{-K}) \;=\; \Pr(X^a_K = 1 \mid \mathbf{x}_{-K}).$$

For conditionally independent judges, this simplifies to: either $\Pr(Y = t{-}1) = 0$ (no pivotal configuration exists) or $\eta_{K'}(a) = \eta_K(a)$.

Proof. From the proof of Theorem 7, the ASR difference is:

$$\begin{aligned}\Delta &= \mathrm{ASR}(\mathcal{P}', a) - \mathrm{ASR}(\mathcal{P}, a) \\ &= \sum_{\mathbf{x}_{-K}:\, s=t-1} \Pr(\mathbf{X}^a_{-K}{=}\mathbf{x}_{-K}) \left[\Pr(X^a_{K'}{=}1 \mid \mathbf{x}_{-K}) - \Pr(X^a_K{=}1 \mid \mathbf{x}_{-K})\right].\end{aligned}$$

Each summand is non-negative (pointwise dominance ensures the bracket is ≥ 0; the probability weight is ≥ 0). The sum vanishes iff every summand with positive weight has a zero bracket. Under conditional independence, $\Pr(X^a_K = 1 \mid \mathbf{x}_{-K}) = \eta_K(a)$ is constant across realizations, so the condition collapses to $\eta_{K'}(a) = \eta_K(a)$ or the event $\{Y = t{-}1\}$ having zero probability. □

Remark. *The condition* $\Pr(Y = t{-}1) = 0$ *is degenerate: it requires all* $\eta_j \in \{0, 1\}$ *with the count of* $\eta_j = 1$ *never equaling* $t{-}1$*. In all non-degenerate panels, the monotonicity is* strict *whenever* $\eta_{K'} > \eta_K$ *under some attack.*

Proposition 9 (Panel ASR Bounds)**.** *For a panel of* K *conditionally independent judges with ordered vulnerabilities* $\eta_{(1)} \leq \cdots \leq \eta_{(K)} = \eta_{\max}$ *under a fixed attack:*

$$\prod_{j=K-t+1}^{K} \eta_{(j)} \;\leq\; \mathrm{ASR} \;\leq\; \sum_{k=t}^{K} \binom{K}{k} \eta_{\max}^k \, (1 - \eta_{\max})^{K-k}.$$

Proof. Lower bound. The event that the t most vulnerable judges $(J_{(K-t+1)}, \ldots, J_{(K)})$ all fail implies panel compromise. By independence:

$$\mathrm{ASR} \;\geq\; \Pr\big(\textstyle\bigcap_{j=K-t+1}^{K} \{X_{(j)} = 1\}\big) \;=\; \prod_{j=K-t+1}^{K} \eta_{(j)}.$$

Upper bound. Since $\eta_j \leq \eta_{\max}$ for all j, replacing each η_j with $\eta_{\max}$ increases ASR by Lemma 6 (applied iteratively to each judge). The resulting panel has i.i.d. Bernoulli($\eta_{\max}$) failures, giving the binomial tail. □

Remark (Scaling Regimes)**.** *For small* $\eta_{\max}$ *and fixed* K*, the leading-order behavior is:*

- *If the top* t *vulnerabilities satisfy* $\eta_{(j)} = \Theta(\eta_{\max})$ *for* $j \geq K{-}t{+}1$*:*

  $$\mathrm{ASR} \;=\; \Theta\big(\eta_{\max}^t\big).$$

  Majority vote provides a t*-th power suppression relative to the individual maximum.*

- *If the vulnerability distribution is* heavy-tailed *(one judge much weaker than the rest): the lower bound* $\prod \eta_{(j)}$ *is small, but the adversary can still exploit correlated failures (attack transfer) to push ASR closer to* $\binom{K}{t}\eta_{\max}^t$*.*

- *For* $K = 3$*,* $t = 2$*:* $\mathrm{ASR} = \Theta(\eta_{\max}^2)$ *in the small-vulnerability regime.*

6 Corollary: Panel Design

Corollary 10 (Robustness-First Design Principle)**.** *Under the adaptive adversary model of Definition 3, the optimal panel selection strategy is* robustness-first*:*

Stage 1*: *Robustness screening. *From the candidate pool, retain only judges satisfying* $\eta_j \leq \tau$ *for a robustness threshold* τ*. By Theorem 7, replacing any judge with* $\eta_j > \tau$ *by one with* $\eta_j \leq \tau$ *weakly reduces panel ASR.*

Stage 2: ***Diversity maximization.*** *Among the retained judges, select the K-judge panel that maximizes effective independence K_{eff} (or any suitable diversity measure). This provides residual protection against broad-spectrum attacks without introducing weak links.*

The ordering is essential: diversity maximization without robustness screening may increase $\eta_{\max}$ via the diversity-capability tradeoff, which by Theorem 7 weakly increases panel ASR.

Proof. Stage 1 is a direct application of Theorem 7: if J_K has $\eta_K > \tau$ and J'_K has $\eta_{K'} \leq \tau < \eta_K$, and J'_K satisfies pointwise dominance of J_K over J'_K (i.e., J_K is pointwise more vulnerable than J'_K), then $\text{ASR}^*(\mathcal{P}) \geq \text{ASR}^*(\mathcal{P}')$. The robustness constraint $\eta_j \leq \tau$ for all panel members ensures $\eta_{\max} \leq \tau$, bounding the dominant predictor.

Stage 2 is a secondary optimization: given $\eta_{\max} \leq \tau$, the residual variance in panel ASR is determined by inter-judge correlations and the specific attack-type vulnerability distribution. Maximizing K_{eff} provides protection in the regime where attacks exploit shared biases (broad-spectrum attacks), as the SDPI bound gives $\text{ASR} \leq K_{\text{eff}} \cdot \eta_{\max}$.

The ordering matters because the diversity-capability tradeoff ($r \approx -0.78$, $p < 0.001$ empirically) implies that panels selected for maximal diversity tend to include weaker judges. By Theorem 7, the resulting increase in $\eta_{\max}$ dominates any benefit from increased independence. □

7 Connection to Empirical Results

Remark (Empirical Validation)**.** *Theorem 7 makes a testable prediction: in a multivariate regression $\log(\text{ASR}) \sim \beta_1 \log(\eta_{\max}) + \beta_2 \log(K_{\text{eff}})$, $\eta_{\max}$ should be a significant predictor ($\beta_1 > 0$) while K_{eff} should be weak or non-significant ($\beta_2 \approx 0$), because the theorem establishes monotonicity in $\eta_{\max}$ but not in K_{eff}.*

Across 12 experimental conditions (2 benchmarks × 6 attack types):

- *$\eta_{\max}$ is significant (BH-FDR $q < 0.05$) in 6/12 conditions (50%), with 3/12 surviving Bonferroni correction.*
- *K_{eff} is significant in only 1/12 conditions (8%). Residualized K_{eff} (orthogonalized against $\eta_{\max}$, VIF $= 1.04$) confirms this is not a collinearity artifact.*
- *The sole exception (ARC × sycophancy, $\beta_2 = -2.85$, $p = 0.003$) corresponds to the one attack type that exploits a* shared *judge bias rather than targeting individual vulnerabilities—the "BFT-like" regime where faults are approximately random.*

This pattern is precisely what Theorem 7 predicts: monotonicity in $\eta_{\max}$ drives the primary effect, and K_{eff} matters only when the adversary cannot selectively target the weakest judge.

Remark (When Diversity Helps)**.** *Theorem 7 does* not *claim diversity is useless. It establishes that* reducing $\eta_{\max}$ weakly dominates increasing K_{eff} *as a defense strategy. Diversity retains value in three regimes:*

(i) ***Controlled*** *$\eta_{\max}$**:** When all judges exceed a robustness threshold (Corollary 10, Stage 2), higher K_{eff} reduces the ASR ceiling via the SDPI bound $\text{ASR} \leq K_{\text{eff}} \cdot \eta_{\max}$.*

(ii) ***Broad-spectrum attacks:*** *When attacks exploit biases shared across judges (sycophancy, position bias), failures approximate random faults and the BFT analogy holds.*

(iii) ***Non-adversarial settings:*** *For calibration, bias reduction, and inter-rater agreement, diversity provides standard statistical benefits unrelated to adversarial robustness.*